

STS EEG Screening Server Update

Global STD fix + image cascades — September 2026

Four steps on each server. Do the development server first; when its results look right, do production the same way. Everything is a file copy through your normal Remote Desktop session — no scripts required.

What this changes

- **Global STD** on page 1 of every Brain Panel currently prints 1000.00 with z 2642.71. After the update it reads about 2.5, inside the normal range. No other metric moves — verified 47 of 48 identical on a real study.
- **An image cascade** (<study>.imagecascade.pdf, about 13 MB, one page pair per ICA component) is written next to every panel. Each study now takes about 8 minutes to process instead of under a minute.
- **Disk:** about 14 MB per study. 1,000 studies use about 15 GB of the 100 GB free. The new watchdog prints a LOW DISK SPACE warning under 10 GB free.

The files

On your workstation, open this folder in Explorer:

```
C:\BrainPanel\clean_eeg_project_2025\deploy\server-2026-09\staged
```

File	Goes into	What it does
files\edftotextbynameplotproc.py	project\files\	Global STD fix
files\edftotextbycommandplotproc.py	project\files\	turns the cascade on
files\Montage_6.py	project\files\	cascade rendering
files\dummy_gui.py	project\files\	cascade head maps
files\create_report_pdf.py	project\files\	cascade screen capture
tomwatchdog_serialized.py	project folder (next to module7.py)	new watchdog — one study at a time

“Project folder” means the directory on the server that contains module7.py and tomwatchdog.py.

On the server

In your Remote Desktop session to the server.

1. Back up the five files being replaced

In the project's `files\` folder, make a new folder called `backup-2026-09` and copy these five files into it:

```
Montage_6.py
create_report_pdf.py
dummy_gui.py
edftotextbycommandplotproc.py
edftotextbynameplotproc.py
```

2. Paste the new files in

- Copy the five `.py` files from `staged\files\` on your workstation and paste them into the project's `files\` folder on the server. Say **Yes** to overwrite.
- Copy `tomwatchdog_serialized.py` and paste it into the project folder itself, next to `module7.py`.

3. Swap the watchdog

In the console window where `tomwatchdog.py` is running, press **Ctrl-C**. Then, in that same window, start the new one:

Development server:

```
py tomwatchdog_serialized.py "c:/app/STSEEGScreening/Source/Practitioners"
```

Production server:

```
py tomwatchdog_serialized.py
```

It prints `monitor dir:` and `interpreter: first` — glance at both. The production path is built in; the development server needs its path given because the two servers watch different folders.

It must run in that console window — not as a service, not from a scheduled task. The cascade captures the screen, and only that session has one.

4. Upload one study and look at the result

Wait about 8 minutes. A fullscreen window flashes roughly 19 times while the cascade is captured — leave it alone; don't cover it or minimize the Remote Desktop window. Then check the files next to the EDF:

Dev	Prod	Check
☐	☐	.icale.rep.pdf — page 1, the Global STD row reads about 2.5 , not 1000.
☐	☐	.imagecascade.pdf exists, about 13 MB.
☐	☐	Open the cascade and page through it: component graphs on every page, no black pages .
☐	☐	Page 1 again: the four Moment 3 rows are in the hundreds to low thousands.

If the Moment 3 rows are in the **tens of thousands**: that server has numpy 2.x. Also paste staged-optional\process\detect_artifact.py into the project's process\ folder and re-run the study. This is a one-time check per server.

That is the whole deployment. Production is the same four steps, starting the watchdog with no argument, in the console session that is kept logged in from the development server.

If something is wrong

1. Copy the five files from backup-2026-09 back into files\.
2. Ctrl-C the new watchdog.
3. Start the old one: `py tomwatchdog.py`

Nothing in this update touches data files, the reference database, or any existing output, so this is a complete undo.

Worth knowing

- **Why the watchdog changes:** the old one starts a new process for every upload, with no limit. Two cascades rendering at once each capture the other's window, and the result is wrong with no error. The new one processes one study at a time, waits for an upload to finish writing before starting, and kills anything that runs past 25 minutes.
- **Development server only:** its screen is live only while you are connected. Stay connected for the 8-minute test in step 4. Production is unaffected — its session is held by the console logged in from dev.
- **Screen saver / lock screen:** keep both off on any machine rendering cascades. A lock screen firing mid-render produces black pages.
- **Older panels:** every panel issued before this update carries the bogus Global STD row. Nothing else on them is wrong; their out-of-bounds count is one too high.
- **Longer documents:** README.md in the same folder explains each change; DEPLOY_STEPS_DEV.md adds verification scripts for pinning down a problem if step 4 looks off.